Pranshu Rastogi

Senior Data Scientist — Finance · Agentic AI Systems · Applied NLP Research

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SUMMARY

Senior Data Scientist with 4+ years shipping production machine learning and agentic AI in financial crime — promoted after **cutting fraud losses 40%**, owning outcomes end-to-end rather than just building models. Curious and self-driven, I stay deliberately current: **3× ACL/NeurIPS-recognised** (Best Paper @ ACL BEA 2025; Bronze @ NeurIPS 2025) and a **Kaggle Expert (Top 2%)**, earned alongside a full-time role. Known for initiative, resilience on open-ended problems, and a pragmatic bias for action — plus the stakeholder communication to turn model behaviour into decisions non-technical partners can act on.

SKILLS

Agentic AI & LLMs: Multi-agent orchestration & tool-use (ReAct / Reflection / Plan-and-Execute), RAG & context engineering, fine-tuning (LoRA/QLoRA, RLHF/DPO), LangChain, LlamaIndex, CrewAI, GPT-4, Claude, Gemini APIs

ML / NLP / Deep Learning: Python, PyTorch, HuggingFace Transformers, Sentence-BERT, DeBERTa, bi-/cross-encoder retrieval, XGBoost, LightGBM, SHAP, Computer Vision (LayoutLM, CNNs)

MLOps & Data: GCP, Apache Airflow, SQL, feature stores, model deployment & monitoring, drift detection (PSI / KS-test / chi-squared), champion/challenger retraining, A/B testing & shadow deployment

Domain & Collaboration: Fraud / FinCrime ML, AML & sanctions screening, model explainability, stakeholder communication, cross-functional delivery

EXPERIENCE

Senior Data Scientist, Tide Bank — London, UK (Remote) **2024 – Present**

- **Cut total fraud losses 40%** by designing and productionising dual-lens fraud models that score APP-fraud victims and perpetrators as independent risk problems — owned end-to-end from problem framing through deployment and live monitoring. **Skills:** *XGBoost, LightGBM, behavioural graph features, SHAP*
- **Eliminated 95% of false-positive sanctions alerts** with zero loss of screening coverage, by building a real-time agentic screening service over live OFAC, UN, EU and HMT delta feeds — automating AML/PEP checks on every transaction with no human in the loop. **Skills:** *Agentic reasoning loops, LLM entity disambiguation, fuzzy-NER*
- **Prevented £800K in fraud losses and raised detection precision 20%** on fraudulent invoices, by building a multi-agent document-understanding engine that outputs ranked evidence reports — catching fabricated invoices every rule-based control had missed, while cutting investigator review time. **Skills:** *LayoutLM, LLM document understanding, CoT reasoning agents*

Data Scientist, Tide Bank — London, UK (Remote) **Aug 2022 – 2024**

- **Saved members £2M annually** in suspicious-activity losses by building Tide's first AML/SAR prevention model — now scoring **15M+ transactions per month** in production. **Skills:** *Gradient boosting, transaction-network features, GCP, Airflow feature store, SHAP*
- **Cut model retraining cycles from 1 month to 1 week** by designing a self-healing champion/challenger MLOps framework and resolving root-cause feature-store inconsistencies — removing manual toil and silent model degradation. **Skills:** *PSI / KS-test / chi-squared drift detection, automated reporting*

PUBLICATIONS & AWARDS (Google Scholar)

Pranshu Rastogi @ FinMMEval 2026: Systems for Tasks 1 and 2 — Prompt-Engineered Gemini for Multilingual and Cross-Lingual Financial Reports QA — Multilingual financial QA with Gemini 3.x, no fine-tuning: structured prompting, cost-gated self-consistency voting, tier-aware routing. ***Outstanding Paper Award.*** *FinMMEval Lab @ CLEF 2026.* Rastogi P. [Code] [Certificate]

NeurIPS 2025 — CURE-Bench Medical AI (Harvard Med / MIT / CZI) — **Bronze · 6th globally.** Confidence-weighted ensemble of frontier LLMs with multi-step agentic reasoning; invited to the NeurIPS 2025 award ceremony (San Diego, Dec 2025); paper in preparation. [Certificate]

Phaedrus at BEA 2025 Shared Task — Tutor identity classification & actionability evaluation in math tutoring dialogues. ***Best Paper Award, Track 5.*** *BEA Workshop @ ACL 2025.* Rastogi P., Tiwari R. [Certificate]

fact check AI at SemEval-2025 Task 7 — Multilingual/crosslingual claim retrieval, bi-encoder + cross-encoder ensembles. ***5th (cross-lingual) / 10th (monolingual) globally.*** *SemEval Workshop @ ACL 2025.* Rastogi P.

Extracting Software Mentions & Relations using Transformers and LLM Synthetic Data — NER + relation extraction via LLM-generated augmentation. *SDP Workshop @ ACL 2025.* Rastogi P., Tiwari R.

Multi-Stage Training for Abusive Comment Detection in Indic Languages — Ensemble pipeline for abuse detection across 13+ Indic languages. *Accepted for presentation, IEEE BigMM 2021.* Rastogi P., Mathur M., Ramaneswaran S., Mohan K.

Glaucoma Detection and Segmentation: A Review and Study — Survey of deep-learning methods. *IEEE INDICON 2020.* Vasudeva J., Yadav J., Rastogi P.

PROJECTS

BTC Mining Fees Optimization via RL — Framed miner transaction selection as a knapsack RL problem with a custom Gym env and reward shaping; PPO beat baseline by **+18.4%** (3.44M vs 2.91M fees), DQN by +6.5%. **PPO, DQN, Stable-Baselines3, PyTorch**

Speaker Diarization & Source Separation — Unified UIS-RNN, a custom d-vector Speech Embedder, and a VAD pre-filter into one end-to-end pipeline; reduced diarisation error on TIMIT. **UIS-RNN, PyTorch, VAD**

News Summarization & Topic Prediction — Fine-tuned TinyBERT for extractive summarisation + topic classification, served via Flask + Streamlit at **<200ms/article**. **TinyBERT, HuggingFace, Flask, Heroku**

EDUCATION

Jaypee Institute of Information Technology — B.Tech, Computer Science (CGPA 8) **Jul 2018 – Apr 2022**